

Yuchen Hou

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DoB: June 2003



EDUCATION BACKGROUND

National University of Singapore

Master of Science in Electrical Engineering

Singapore

Aug 2025 – Now

- **Research Experience:** Core member of the **NUS CORE Lab (Control, Robotics & Artificial Intelligence)**, focusing on **Embodied AI** and autonomous robotic decision-making.
- **Research Focus:** Perception and planning for aerial robots based on **Vision-Language Navigation (VLN)**, and the application of **End-to-end Reinforcement Learning** in complex 3D environments.

Nanjing University of Aeronautics and Astronautics

Bachelor of Engineering in Electronic and Communication Engineering

Nanjing, CHINA

Sept 2021 – Jul 2025

- **GPA:** 88/100
- **Key Courses (With Scores):** Advanced Mathematics (96), Probability Theory and Mathematical Statistics (97), Random Signal Analysis (97), Microwave Technology and Antenna (97), Fundamentals of Communication (95), Complex Function (99).

PROJECT EXPERIENCES

LangGap: Diagnosing and Closing the Language Gap in Vision-Language-Action Models

First Author NUS CORE Lab

Oct 2025 – Jun 2026

- **Status:** **Accepted to IROS 2026.**
- **Preprint:** [arXiv:2603.00592](https://arxiv.org/abs/2603.00592)
- Discovered that state-of-the-art VLA models achieve high benchmark scores but **ignore language instructions**, revealing a gap between performance and true understanding.
- Designed systematic **semantic perturbation** evaluation framework to expose language grounding failures.
- Proposed **multi-task same-scene training** approach and constructed augmented dataset for fine-tuning.

Zero-Shot Indoor Object Navigation for Autonomous Drones Using Dual-Policy Reinforcement Learning

Team Member NUS CORE Lab

Jun 2025 – Jun 2026

- **Status:** **Accepted to IROS 2026.**
- **Preprint:** [arXiv:2601.15614](https://arxiv.org/abs/2601.15614)
- Developed a dual-mode visual navigation system based on **end-to-end Reinforcement Learning**, utilizing **RGB-D sensor** input.
- Engineered the policy to autonomously switch between **'Exploration'** and **'Target-driven Navigation'** modes based on target visibility.
- Trained the agent from scratch to master a **discrete action space**, including complex maneuvers like take-off, navigation, and landing.

Vision-Language Navigation on Autonomous Drone

Team Member NUS CORE Lab

Jun 2025 – Feb 2026

- **Status:** **Nearing completion.**
- Built a robust pipeline to generate various **3D paths** in the **Habitat simulator**.
- Overcame the challenges of the simulator initially designed only for ground robots by designing a robust **3D navigation algorithm** and obstacle detection method.
- Trained a strong and general policy for **drone navigation**.

Frontier Based Autonomous Exploration Vehicle Design with ROS2 and Lidar

Team leader NUAA

Sep 2024 – Apr 2025

- Led a team of two members to develop an autonomous exploration system using **ROS2** and **LiDAR** technology.
- Implemented **SLAM** algorithms including **Cartographer** and **Navigation2**, achieving real-time mapping and path planning.
- Integrated **YOLOv11** for object detection.
- Successfully deployed the complete system on an **embedded hardware platform**.

WORK EXPERIENCE

Honor Device Co., Ltd.

Shanghai, CHINA

VLA Algorithm Engineer (Intern)

Jun 2026 – Now

- Working on Honor's self-developed **Omega humanoid robot**: teleoperation data collection, replay, training, and on-robot deployment.
- Conducting **online RL** algorithm research for **Vision-Language-Action (VLA)** models on custom manipulation tasks:
 - Implemented **smooth human-teleoperator takeover** during autonomous rollouts, including the autonomy–teleoperation **transition state machine**.
 - Built the intervention-driven **online-learning data pipeline**: on-robot data recording, full local disk persistence, and automatic upload to the development server.
 - Closed the loop over a **distributed learning architecture**: onboard **AGX/NX** boards (topic publishing, command execution, data collection), an **x86 host** as the **Actor** (model inference, streaming actions to the robot), and a development server as the **Learner** (data aggregation, training, checkpoint dispatch) — a complete **online SFT / online RL** pipeline.
 - Researching **RECAP**-style ($\pi_{0.6}$) **offline RL** combined with **noise-space fast adaptation** (DSRL-like): learning corrective noise from human-intervention data to rapidly imitate takeover behaviors before a new checkpoint finishes training, substantially raising success rates on tasks uncovered by offline training, and further adapting the algorithm to **loco-manipulation** tasks.

Huawei Norbert Wiener Research Center

Singapore

Research Intern

Mar 2026 – May 2026

- Developed backend **deep learning** models for **short-range radar** signal processing.
- Conducted research on downstream tasks including **human pose estimation** and **indoor mapping/SLAM** from radar data.

Movel.AI

Singapore

Robotics Software Intern

Dec 2025 – Jan 2026

- Validated autonomous **navigation and control systems** for forklift robots in industrial environments.
- Conducted on-site **factory deployment** and testing in real warehouse settings.
- Integrated robotics software with **fleet management systems**.
- **Tech Stack**: **ROS2**, Navigation Stack, Sensor Integration.

SKILLS & AWARDS

Applied Skills

- **Programming & Systems**: **Python** (Proficient), **Linux (Ubuntu)**, Shell scripting.
- **AI Frameworks & Simulation**: **PyTorch** (DRL policies), **Habitat**, **AI2-THOR** (Complex 3D environments).
- **Robotics Development**: **ROS2** (Nodes, Comm), **SLAM** (Cartographer, Nav2), **YOLO** integration.

Research Capabilities

- **Academic & Theory**: **Sim-to-Real Transfer**, Embodied AI Architectures, **RL Algorithms**.
- **Literature & Writing**: Tracking top-tier conferences (**CVPR/IROS/ICRA**), Academic Paper Writing.

Languages

- **English (Proficient)**: IELTS Overall 7.0 (Reading: 8.5, Writing: 7.5, Listening: 6.5, Speaking: 6.0).
- **Mandarin**: Native Speaker.

Awards

- Second-class Scholarship (2024) and Third-class Scholarship (2021, 2022, 2023) at Nanjing University of Aeronautics and Astronautics (NUAA).